

**Module 4 - R Practice**

**HYPOTHESIS TESTING USING T-TESTS IN R**

**ALY6010 20185**

**Probability Theory and Intro Statistics**

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**Abstract**

This report applies inferential statistical methods to two separate datasets using R. In Part 1, a Welch two-sample t-test was conducted to evaluate whether male and female cats from the MASS package cats dataset differ in mean body weight. In Part 2, a paired t-test was used to determine whether a meditation workshop improved sleep quality scores among ten students. Results indicated that male cats weigh significantly more than female cats on average ( $t(136.84) = 8.7095$ ,  $p < .001$ ), and that the meditation workshop was associated with a statistically significant improvement in sleep quality scores ( $t(9) = 1.9481$ ,  $p = .042$ ). The selection of each test was justified based on the structure of the data and the nature of the research questions. Findings are discussed in the context of both statistical and practical significance.

## **Hypothesis Testing Using T-Tests in R**

### **Introduction**

This report presents the results of two hypothesis tests conducted in R using the `t.test` function. The first test examines whether male and female cats differ in mean body weight using the cats dataset embedded in the MASS package. The second test evaluates whether a meditation workshop improved sleep quality scores among a sample of ten students. For each analysis, the appropriate test procedure is identified and justified, hypotheses are stated, descriptive statistics are reported, R output is presented, and findings are interpreted in the context of the research question. All tests were conducted at a significance level of  $\alpha = .05$  unless otherwise stated.

### **Part 1: Do Male and Female Cats Have the Same Body Weight?**

#### **Research Question**

The first analysis asks whether male and female cats in the cats dataset have the same mean body weight. This question is answered using a two-sample t-test comparing the mean body weight in kilograms between the two sex groups.

### **Hypotheses**

The null and alternative hypotheses for this test are stated as follows:

$$H_0: \mu_M = \mu_F$$

$$H_1: \mu_M \neq \mu_F$$

The null hypothesis states that male and female cats have the same mean body weight in the population. The alternative hypothesis states that the two groups differ in mean body weight. This is a two-tailed test because the research question asks whether a difference exists in either direction without specifying which sex is expected to weigh more (Bluman, 2019).

### **Test Selection and Justification**

A two-sample Welch t-test with unequal variances was selected for this analysis. The male and female cat groups represent two independent samples because the body weight measurement of any male cat has no relationship to the body weight measurement of any female cat in the dataset. This satisfies the independence assumption that Bluman (2019) identifies as a foundational requirement of the two-sample t-test. The population standard deviations are unknown, and the two groups show substantially different sample standard deviations (0.4675 for males versus 0.2740 for females), meaning the equal variance assumption required by the pooled t-test cannot be justified. Kabacoff (2015) notes that setting `var.equal = FALSE` in R's `t.test` function applies Welch's correction to the degrees of freedom, which produces more reliable inference when group variances differ. This makes

the Welch t-test the more defensible choice when working with real data where variance equality is difficult to verify in advance.

### **Descriptive Statistics**

Table 1 presents the descriptive statistics for body weight by sex group.

*Table 1*

*Descriptive Statistics for Cat Body Weight by Sex*

| <b>Group</b> | <b>n</b> | <b>Mean (kg)</b> | <b>SD (kg)</b> |
|--------------|----------|------------------|----------------|
| Male         | 97       | 2.9000           | 0.4675         |
| Female       | 47       | 2.3596           | 0.2740         |

The male cat sample contained 97 observations with a mean body weight of 2.9000 kg and a standard deviation of 0.4675 kg. The female cat sample contained 47 observations with a mean body weight of 2.3596 kg and a standard deviation of 0.2740 kg. The observed mean difference between the two groups was 0.5404 kg, with male cats weighing more on average. The notably different standard deviations between the two groups confirm that male body weights are more variable than female body weights, which further justifies the selection of Welch's test over the pooled variance alternative.

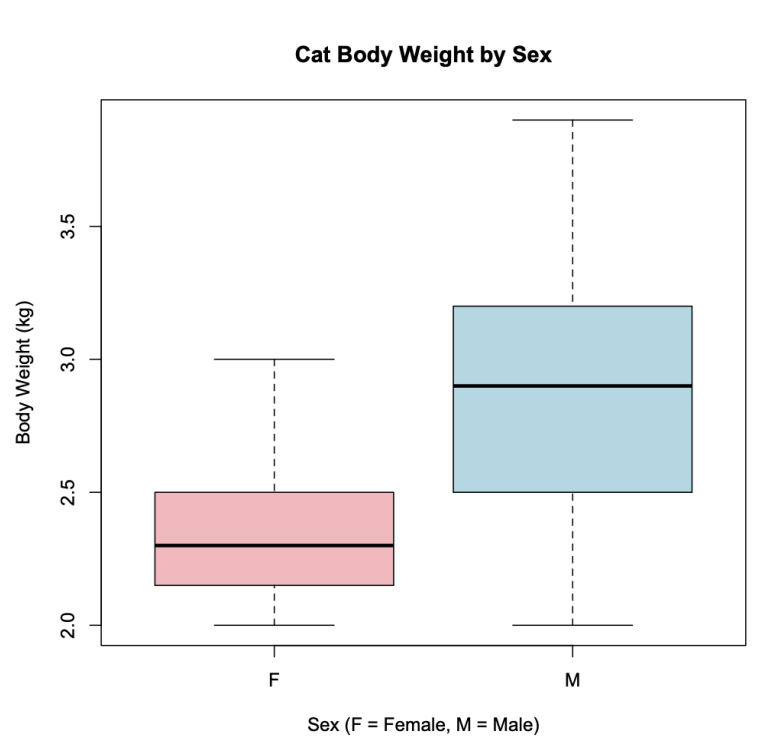
### **Results**

The Welch two-sample t-test produced the following result:

$$t(136.84) = 8.7095, p < .001$$

The 95% confidence interval for the difference in means was [0.4177, 0.6631] kg.

## Data Visualization



*Figure 1*

### *Boxplot of Cat Body Weight (kg) by Sex*

*Note.* Boxes represent the interquartile range. The horizontal line within each box represents the median. Whiskers extend to 1.5 times the interquartile range. Outliers are not displayed.

## Interpretation

Because the p-value of  $8.831e-15$  is far below the significance level of  $\alpha = .05$ , the null hypothesis is rejected. There is extremely strong statistical evidence that male and female cats differ in mean body weight in the population represented by this dataset. The 95% confidence interval of  $[0.4177, 0.6631]$  kg does not contain zero, which further confirms that a difference of zero is not a plausible value for the true population mean difference. Male cats weigh on average approximately 0.54 kg more than female cats.

This finding is both statistically significant and practically meaningful. For animals of this size, a difference of over half a kilogram represents a substantial proportion of total body weight. Figure 1 visually supports this conclusion, showing that the distribution of male body weights is both higher and more spread out than that of female cats, which is consistent with the larger standard deviation observed in the male group. As Bluman (2019) cautions, statistical significance alone does not confirm practical importance, but in this case the magnitude of the difference alongside the extremely small p-value together provide strong and meaningful evidence of a real sex-based difference in body weight among domestic cats.

## **Part 2: Does Meditation Improve Sleep Quality?**

### **Research Question**

The second analysis evaluates whether a one-week meditation workshop improved sleep quality scores among ten students. Sleep quality was measured on a scale of 0 to 10 in the week before and the week after the workshop, with higher scores indicating better sleep quality.

### **Hypotheses**

The null and alternative hypotheses are stated as follows:

$$H_0: \mu d = 0$$

$$H_1: \mu d > 0$$

where  $d$  represents the within-subject difference in sleep quality score calculated as the after score minus the before score for each student. The null hypothesis states that the meditation workshop has no effect on sleep quality, meaning the mean difference across students equals zero. The alternative hypothesis states that sleep quality scores are higher after the workshop

than before. This is a one-tailed test in the positive direction because the researchers specifically claimed that meditation improves sleep quality, which constitutes a directional hypothesis (Bluman, 2019).

### **Test Selection and Justification**

A paired t-test was selected rather than an independent two-sample t-test. The same ten students provided sleep quality measurements both before and after the workshop, meaning the two sets of observations are not independent of each other. Each student's pre-workshop score and post-workshop score are directly linked because they come from the same individual. As Bluman (2019) explains, when data consist of two measurements taken from the same subjects under different conditions, the paired t-test is the appropriate procedure because it isolates the within-subject change and removes individual differences that would otherwise inflate the error variance and reduce statistical power. Applying an independent two-sample t-test to these data would treat the observations as unrelated, which would violate the independence assumption and produce less reliable results. Kabacoff (2015) confirms that setting `paired = TRUE` in R's `t.test` function correctly accounts for this repeated measures structure by computing and testing the mean of the within-subject differences directly rather than comparing two separate group means.

### **Descriptive Statistics**

Table 2 presents the descriptive statistics for sleep quality scores before and after the workshop, along with the paired differences.

*Table 2*

*Descriptive Statistics for Sleep Quality Scores Before and After Meditation Workshop*

| Measure                         | Mean | SD     |
|---------------------------------|------|--------|
| Before workshop                 | 6.25 | 1.9295 |
| After workshop                  | 6.87 | 1.2807 |
| Difference (after minus before) | 0.62 | 1.0064 |

*Note.*  $n = 10$  for all measures.

### Results at $\alpha = .05$

The paired t-test produced the following result:

$$t(9) = 1.9481, p = .042$$

The 95% confidence interval for the mean difference was  $[0.0366, \infty)$ .

Because  $p = .042$  is less than  $\alpha = .05$ , the null hypothesis is rejected. There is sufficient statistical evidence at the .05 significance level to support the claim that meditation improves sleep quality. It is important to note, however, that the p-value sits only marginally below the .05 threshold, which means the evidence, while statistically significant, is not strong. This finding warrants careful interpretation and suggests that a larger sample would provide more reliable and conclusive results.

### Results at $\alpha = .10$

At  $\alpha = .10$ , the t-statistic and p-value remain identical at  $t(9) = 1.9481$  and  $p = .042$  respectively, since changing the confidence level does not affect the test statistic or the p-value. The 90% confidence interval for the mean difference is  $[0.1798, \infty)$ , which lies further above zero than the 95% interval. The conclusion does not change at  $\alpha = .10$  because



the p-value of .042 falls below both significance thresholds. The null hypothesis is rejected at both levels and the conclusion that meditation is associated with improved sleep quality holds regardless of which significance level is applied.

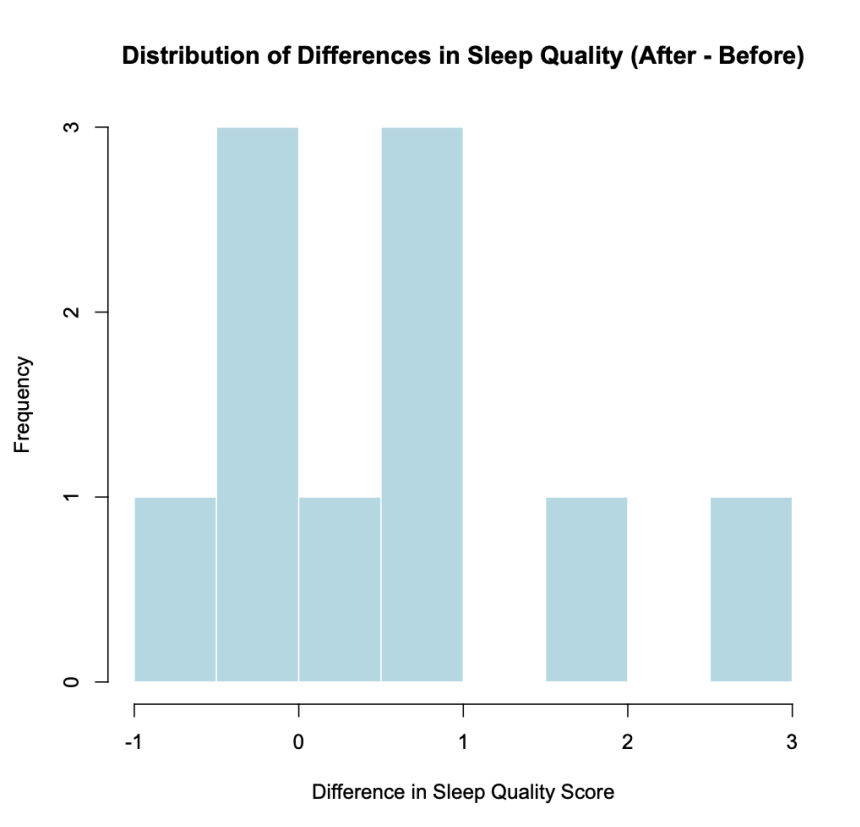
## Data Visualizations



*Figure 2*

*Boxplot of Sleep Quality Scores Before and After the Meditation Workshop*

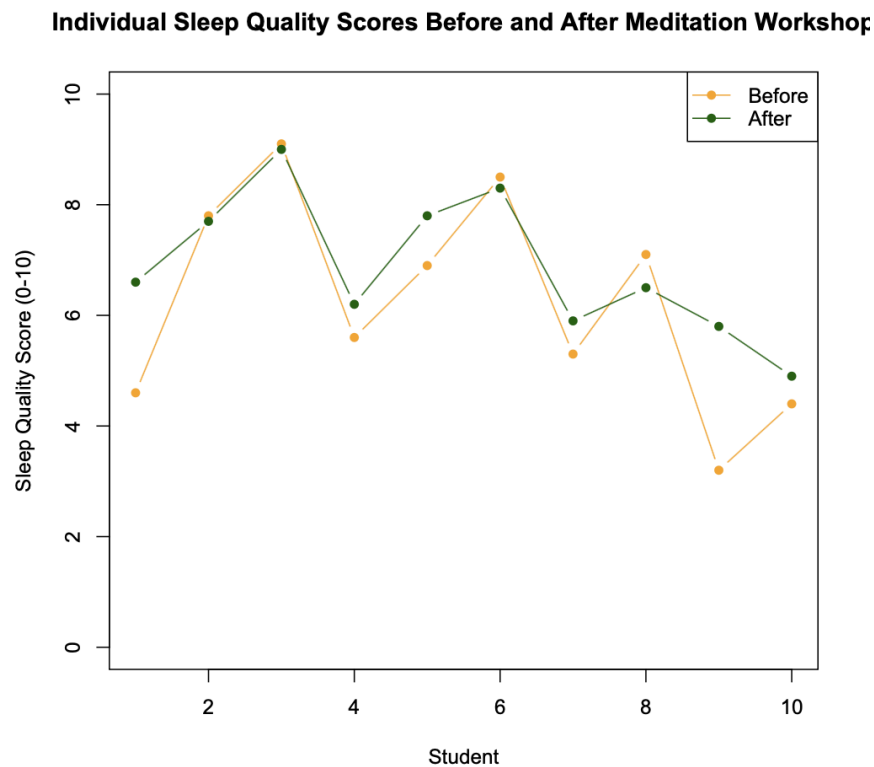
*Note.* Boxes represent the interquartile range. The horizontal line within each box represents the median. Whiskers extend to 1.5 times the interquartile range.



*Figure 3*

*Histogram of Paired Differences in Sleep Quality Scores (After Minus Before)*

*Note.* Each bar represents the count of students whose difference score fell within that interval. A difference greater than zero indicates improvement following the workshop.



*Figure 4*

*Individual Sleep Quality Score Trajectories for Each Student Before and After the Meditation Workshop*

*Note.* Each line connects the before and after scores for a single student. Green lines indicate improvement in sleep quality following the workshop. Red lines indicate a decline.

### **Interpretation**

The paired t-test provides statistically significant evidence that the meditation workshop was associated with improved sleep quality scores among the ten students. The mean score increased from 6.25 before the workshop to 6.87 after the workshop, representing a mean improvement of 0.62 points on the 10-point scale. This conclusion holds at both the .05 and .10 significance levels because the p-value of .042 falls below both thresholds.

The histogram in Figure 3 shows that the distribution of individual differences is largely positive, meaning most students recorded higher sleep quality scores after the workshop than before. This visual pattern is consistent with the direction of the test result and supports the normality assumption underlying the paired t-test, which is the key distributional assumption when working with a small sample of  $n = 10$  (Bluman, 2019).

Figure 4 provides the most informative visualization of the paired data structure. The majority of students show an upward trajectory from before to after the workshop, which confirms the overall pattern of improvement. A small number of students recorded lower scores after the workshop, which contributes to the variability in the differences and explains why the p-value, while significant, sits close to the .05 boundary rather than well below it.

Bluman (2019) draws a clear distinction between statistical significance and practical significance, and this distinction is particularly relevant here. A mean improvement of 0.62 points on a 10-point scale is modest in practical terms, and whether this magnitude represents a clinically or practically meaningful change in sleep quality would require additional domain judgment beyond what the statistical test alone can provide. Furthermore, the sample of ten students is small, which limits the generalizability of these findings to broader populations. Future studies with larger and more diverse samples, along with controlled conditions that account for potential confounding factors such as academic workload, lifestyle changes, and seasonal variation in sleep patterns, would provide stronger and more generalizable evidence of the relationship between meditation and sleep quality.

## **Conclusion**

This report applied two t-test procedures to answer substantive questions using real data in R. In Part 1, a Welch two-sample t-test demonstrated extremely strong evidence that male and female cats differ in mean body weight, with male cats weighing on average 0.54 kg more

than female cats ( $t(136.84) = 8.7095$ ,  $p < .001$ ). In Part 2, a paired t-test provided statistically significant but modest evidence that the meditation workshop was associated with improved sleep quality scores among ten students ( $t(9) = 1.9481$ ,  $p = .042$ ). The selection of each test was driven by the structure of the data. Independence between groups justified the Welch test in Part 1, and the within-subject repeated measures design justified the paired test in Part 2. Using an independent sample test for Part 2 would have been methodologically incorrect given that the same students were measured twice. Taken together, these analyses demonstrate that selecting the appropriate hypothesis test based on data structure and research design is fundamental to drawing valid and defensible statistical conclusions from real data.

## References

- Bluman, A. G. (2019). *Elementary statistics: A step by step approach* (10th ed.). McGraw-Hill.
- Kabacoff, R. (2015). *R in action: Data analysis and graphics with R* (2nd ed.). Manning Publications.
- Venables, W. N., & Ripley, B. D. (2002). *Modern applied statistics with S* (4th ed.). Springer.